Beyond Visual Plausibility: Consequence-Aware Action Verification for 3DGS Robot Digital Twins

Anonymous CVPR 2026 submission

Paper ID ****

Abstract

3D Gaussian Splatting (3DGS) is increasingly used as a visual backbone for embodied AI, supporting photorealistic robot digital twins, demonstration generation, and action-conditioned scene prediction. These systems provide visually realistic and spatially consistent robot scenes and demonstrations, while reliable execution additionally requires candidate actions to realize the intended physical task consequence. This paper studies candidate action consequence verification in 3DGS robot digital twins: given a reconstructed scene, a task goal, and a candidate action, the verifier estimates whether the action will satisfy the task predicate after physical interaction. As a motivating diagnostic, we construct a Same-Visual-Different-Physics setting that keeps the rendered observation, object pose, camera, robot trajectory, and kinematic label fixed while varying hidden physical properties such as friction, mass, center of mass, and contact geometry. The resulting physical outcomes diverge under identical visual labels, exposing a gap between visual/kinematic plausibility and physical consequence consistency. This observation motivates a physics-aware action-effect verification layer for 3DGS robot demonstrations and robot digital twins.

1. Introduction

3D Gaussian Splatting (3DGS) provides embodied AI with a high-fidelity, reconstructable, and renderable 3D scene representation. It recovers photorealistic appearance from real images and renders novel views efficiently [10]. This capability allows robot systems to build digital twins from real environments and to obtain broader variation in viewpoint, illumination, object pose, and scene configuration beyond limited real-robot data.

Recent work has brought 3DGS into robot simulation, demonstration generation, and future prediction. RoboSplat uses 3DGS to generate visually realistic robot demonstrations from a single expert demonstration [27]. GSWorld combines 3DGS rendering with ManiSkill, SAPIEN, and PhysX to form a photorealistic closed-loop robot simulation environment [8, 23, 20, 21]. RoboGSim, SplatSim, and Re^{3}Sim further advance real-to-sim, sim-to-real, and high-fidelity manipulation data generation [11, 22, 5]. ManiGaussian, GWM, and DreMa use Gaussian representations for dynamic scene modeling and action-conditioned future prediction [14, 13, 2]. These works show that 3DGS has expanded from a static visual representation into a visual backbone for robot data, simulation, and world models.

Robot execution still requires choosing among candidate actions. A visually realistic scene or a kinematically plausible demonstration does not by itself answer the execution-time question: which candidate action will produce the intended task consequence? A grasp can appear aligned with the target object in the image but fail because of contact location, gripper width, friction, or center-of-mass variation. A push can point toward the target region yet produce insufficient displacement because the contact point is offset. A placement can be kinematically reachable but fail due to collision-proxy or surface-closure errors.

This paper distinguishes three levels of plausibility. Visual plausibility describes whether an observation, rendering, or prediction looks realistic. Kinematic plausibility describes whether a trajectory, pose, or object-pose sequence is geometrically feasible. Consequence consistency further requires the physical state after action execution to satisfy the task objective, such as stable grasping, moving the target object, completing a placement, triggering a state change, or satisfying a final-state relation. We study the third level: the task consequence of a candidate action after physical interaction.

Candidate action consequence verification sits after generation or prediction and before execution. Given a 3DGS scene or its derived observation and geometric representation, a task goal, and a set of candidate actions, the verifier outputs a success probability, ranking score, filtering decision, or confidence estimate. Candidate actions may come from a policy, planner, vision-language model, demonstration augmentation module, Gaussian world model, or interactive action proposal. The verifier asks whether these candidates will satisfy the task predicate after execution.

This problem is especially pronounced in 3DGS robot systems. Photometric reconstruction primarily optimizes multi-view image explanation, while Gaussian density does not directly guarantee contactable surfaces, closed geometry, or collision proxies. Hidden physical properties are typically uncertain in a 3DGS representation, including friction, mass, center of mass, contact geometry, control error, and contact response. Visually identical scenes can correspond to different physical worlds; contact-rich manipulation amplifies these differences, making visual and kinematic labels insufficient for certifying final task consequences.

A motivating diagnostic already reveals this gap. In RoboSplat-style demonstrations, the rendered observation and kinematic label can remain unchanged, while physics replay succeeds or fails under object-pose error or hidden physical-property variation. This observation indicates that visual/kinematic success labels provide useful descriptions of generated trajectories, but consequence verification must further reason about physical interaction, hidden properties, and task predicates.

These observations motivate a physics-aware action-effect verification layer for 3DGS robot demonstrations and robot digital twins. Such a layer complements reconstruction, demonstration generation, and action-conditioned prediction by estimating whether a proposed action is likely to satisfy the task predicate before execution, thereby supporting action selection and safety filtering.

2. Related Work

2.1. Visual Scene Representations

Neural scene representations have changed how robot systems acquire real-world appearance. NeRF encodes multi-view images as a continuous radiance field, making novel-view synthesis a learnable 3D representation problem [16]. 3D Gaussian Splatting advances high-quality rendering to an interactive regime through explicit Gaussian primitives and efficient rasterization [10]. This efficiency matters for robotics, where camera viewpoints change with hand-eye motion and training or evaluation requires variation in environment, lighting, object pose, and camera configuration. Recent surveys show that 3DGS has entered robot mapping, navigation, manipulation, and embodied perception [30].

Visual scene representations provide robots with a real-world appearance substrate, but the state variables needed for manipulation are not fully constrained by photometric objectives. Gaussian density can render accurate images without yielding closed surfaces, stable collision proxies, or reliable contact normals. SuGaR and 2DGS improve surface consistency and geometric quality for Gaussian representations [4, 7], illustrating the intermediate layer between renderable appearance and interactable geometry. Candidate action consequence verification sits after this layer: it uses the visual and spatial cues provided by 3DGS while asking whether those cues support the task-level consequence of a contact action.

2.2. Interactive Digital Twins

When 3DGS enters the robot data loop, it takes on a system role beyond scene reconstruction. RoboSplat generates visually realistic manipulation data from Gaussian objects and a single expert demonstration [27]. SplatSim uses Gaussian representations of real scenes for zero-shot sim-to-real transfer of RGB manipulation policies [22]. RoboGSim and Re^{3}Sim further use 3DGS as a high-fidelity carrier for real-to-sim-to-real manipulation data generation [11, 5]. These works mark an important shift: the real world is not only the final test environment, but can also enter training, augmentation, and evaluation through renderable digital twins.

The other side of interactive digital twins is the physics interface. SAPIEN and ManiSkill provide accessible physical environments for articulated-object interaction and large-scale manipulation skill evaluation [23, 20], while physics engines such as PhysX provide contact, constraint, and rigid-body dynamics support [21]. GSWorld connects 3DGS rendering to a closed-loop manipulation simulator, allowing physical states and photorealistic observations to evolve in the same system [8]. This direction now contains key components for generating candidate demonstrations and executing physical rollouts; the decision interface between them still needs to be explicitly defined, namely what consequence confidence a generated or planned candidate action should receive before execution.

2.3. Action-Conditioned Prediction

World models extend robot decision making from current observations to possible futures. ManiGaussian uses dynamic Gaussian Splatting to model future scenes in multi-task manipulation [14]. GWM extends Gaussian representations into scalable manipulation world models [13]. DreMa uses compositional imagination to support imitation learning [2]. More general action-conditioned prediction systems, such as RoboDreamer, Video Prediction Policy, and ChronoDreamer, also show that visual future prediction can provide useful intermediate variables for planning, control, and policy learning [28, 6, 29].

These models enable robots to unfold imagined futures before execution. In visually grounded systems, natural-looking future frames, continuous trajectories, and plausible object motion often serve as important model-quality evidence. Manipulation tasks ultimately require task predicates: whether the object is stably grasped, whether displacement reaches the target region, whether a door or drawer reaches the desired state, and whether slip or collision occurs during contact. Action-conditioned prediction therefore provides visual hypotheses about candidate consequences, while consequence verification must convert those hypotheses into rankable and filterable execution confidence.

2.4. Physics-Aware Gaussian Representations

To narrow the distance between visual scenes and physical interaction, recent work adds dynamics, material properties, and geometric structure to Gaussian representations. PhysGaussian integrates physical dynamics into 3D Gaussians for generative dynamic scene modeling [24]. Physics3D and GaussianProperty estimate physical properties in Gaussian scenes from video diffusion or multimodal model cues [12, 26]. Physically Embodied Gaussian Splatting and Splatting Physical Scenes explore hybrid real-to-sim paths connecting correctable world models, explicit geometry, and real robot data [1, 19].

This trend indicates that 3DGS-based embodied AI is moving from appearance-centric representation toward interaction-aware representation. For robot manipulation, physical properties are still only part of the task consequence. The same friction, mass, or collision geometry can produce different outcomes under different gripper widths, approach directions, controller gains, and task predicates; uncertainty in property estimation can also be amplified by a contact sequence. Candidate action verification therefore needs to combine physical belief, visible geometry, action parameters, and goal predicates, placing property recovery inside action-conditioned consequence reasoning.

2.5. Action Success Prediction

Robot learning has long studied action feasibility and success probability. Dex-Net/GQ-CNN combines depth observations, synthetic grasp metrics, and robust grasp planning to predict grasp success [15]. QT-Opt formulates vision-based grasping as a scalable closed-loop action-value learning problem [9]. For richer interaction tasks, Where2Act learns actionable regions and action directions on articulated objects [17]; Deep Affordance Foresight predicts future affordances for long-horizon tasks [25]; task success classifiers learn task-completion judgments from a small number of real demonstrations [18]. Visual Foresight further shows the connection between action-conditioned prediction and model-based visual control [3].

These works provide a mature robot-learning context for deciding whether an action is worth executing. 3DGS robot digital twins bring this problem into a new representational setting: the input scene comes from photorealistic reconstruction, where appearance error may be small while contact geometry, closed surfaces, collision proxies, and hidden physical properties remain uncertain. Candidate actions may also be proposed by demonstration generation, Gaussian world models, planners, or vision-language models. Candidate action consequence verification can be viewed as action success prediction specialized to 3DGS digital twins, where task predicates, action parameters, and 3DGS-derived geometric/physical uncertainty jointly determine execution confidence.

3. Problem Formulation

We consider candidate action consequence verification in a 3DGS robot digital twin. Let S denote a 3DGS scene or a representation derived from 3DGS observations, geometry, visibility, and uncertainty. S may contain Gaussian primitives, rendered images, multi-view observations, depth or point-cloud estimates, semantic or instance cues, and geometric confidence. Let g denote the task goal, such as grasping a specified object, pushing an object into a target region, placing an object into a container, or reaching a final-state relation. The candidate action set is

$$\mathcal{A} = \{a_1, a_2, \ldots, a_N\},$$

where each a_i is a parameterized grasp, push, place, open/close, or other manipulation action.

The consequence verifier is denoted as V. It receives the scene, task goal, and candidate action, and outputs a consequence score:

$$V(S, g, a_i) \rightarrow r_i,$$

where r_i may represent the probability, ranking score, filtering decision, or confidence that action a_i will produce the intended task consequence. Action selection can be written as

$$a^\star = \arg\max_{a_i \in \mathcal{A}} V(S, g, a_i),$$

and filtering can be written as

$$\mathcal{A}_{\text{valid}} = \{a_i \in \mathcal{A} \mid V(S, g, a_i) > \tau\}.$$

Here, V denotes the consequence-aware verifier under study; its concrete implementation is defined by the method.

The true physical consequence depends on physical and geometric factors that are unobserved or uncertain in 3DGS. Let ξ denote hidden physical properties and execution disturbances, including friction, mass, center of mass, contact geometry, collision-proxy error, control error, and contact response. The terminal state after action execution can be written as

$$s_T = \mathcal{T}(S, a_i, \xi),$$

where $\mathcal{T}$ is the state-transition process in the real world or simulator. The task goal g defines a success predicate ϕ_g, and the true task-consequence label is

$$y_i = \mathbb{I}[\phi_g(s_T) = 1].$$

The verifier aims to estimate y_i under partially observed ξ and $\mathcal{T}$, or to rank candidate actions satisfying the task predicate above failing actions.

This definition differs from visual scoring and kinematic validation. Visual scoring asks whether a rendered or predicted observation looks realistic. Kinematic scoring asks whether a trajectory is feasible under an assumed geometric model. Consequence verification asks whether the terminal state after contact-rich interaction satisfies the task predicate. A candidate action may have high visual plausibility and kinematic plausibility while lacking consequence consistency under hidden physical-property or contact-condition changes.

4. Diagnostic Study: Visual and Kinematic Success Do Not Certify Physical Consequences

This section uses physics replay of RoboSplat-style demonstrations to characterize the gap between visual/kinematic labels and physical task consequences. The diagnostic results provide empirical motivation for consequence-aware verification.

Replay setting	Rollouts	Kinematic label	Physical success
Nominal pose	100	1.0	1.000
Pose perturbation	507	1.0	0.053

Table 1. RoboSplat physical replay under pose uncertainty. The kinematic label is 1.0 for both nominal and perturbed replay, while physical success decreases sharply under pose uncertainty.

4.1. RoboSplat as a Kinematic Demonstration Generator

RoboSplat provides a useful case study for examining the relation between visual demonstration generation and physical execution. It produces visually realistic and spatially consistent robot demonstrations from 3D Gaussian objects [27]. In the original generation pipeline, the object is represented as Gaussian primitives, while collision geometry and contact dynamics are outside the generated object representation. During the grasped phase, object Gaussians follow the end-effector motion by construction, yielding a kinematic label tied to the generated visual trajectory. This property makes RoboSplat an appropriate diagnostic setting for studying how generation-time visual/kinematic labels relate to physical action consequences.

4.2. Physics Replay under Pose Uncertainty

We replay RoboSplat demonstrations in a SAPIEN-PhysX scene with a physical proxy object, using the same Panda robot, controller, and timestep. The zero-offset replay serves as a sanity check: when the physical object is placed at the pose assumed by the generated demonstration, all 100 nominal replays succeed, with a mean maximum lift of 0.150 m. Under object-pose perturbations, physical success decreases sharply. Across 507 perturbed rollouts, the physical success rate is 0.053, while the generation-time kinematic label remains 1.0 for all cases. The failures are dominated by weak grasps, indicating a structured contact mechanism rather than random controller noise.

Table 1 reports the core statistics of this diagnostic. The contact rate remains 1.00 in the perturbation experiment, and failures mainly come from contact quality and grasp stability. As object-pose error increases, physical success drops from 0.73 in the sub-centimeter range to nearly 0 beyond 4 cm. This trend turns the gap between kinematic labels and physical task consequences into a measurable verification problem.

4.3. Same Visual, Different Physics

The Same-Visual-Different-Physics diagnostic removes pose error from the analysis. The rendered

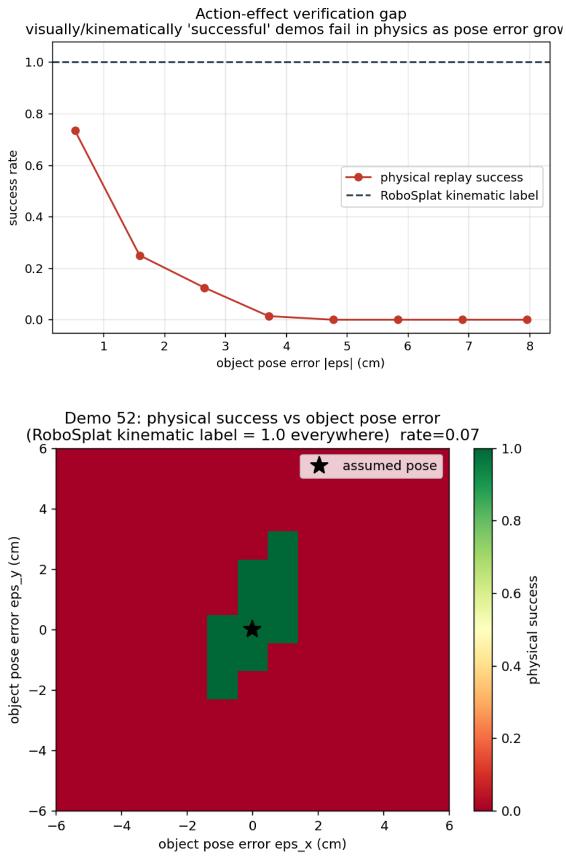

Figure 1. Physics replay under object-pose uncertainty. Replay succeeds when the physical object is placed at the pose assumed by the generated demonstration. Under small object-pose errors, physical success decreases rapidly while the kinematic label remains 1.0. This shows that the generation-time kinematic label does not provide a robust execution certificate under test-time state uncertainty.

3DGS observation, object pose, camera, robot trajectory, and kinematic label are fixed, and only hidden physical properties are varied in the replay simulator. The varied properties include friction, mass, center-of-mass offset, and hidden collision width. Every case retains the same visual/kinematic label, while physical execution outcomes change under more realistic grasp-force conditions.

Table 2 summarizes the results under a more realistic 5 N grasp-force setting. The nominal configuration has a physical success rate of 1.00, indicating that the replay environment, controller, and proxy object support successful execution. After varying hidden physical properties, the success rate is 0.43 for the friction sweep, 0.70 for the mass sweep, 0.64 for center-of-mass offset, and 0.60 for hidden collision width. The main failure modes include slip, weak grasp, and unstable

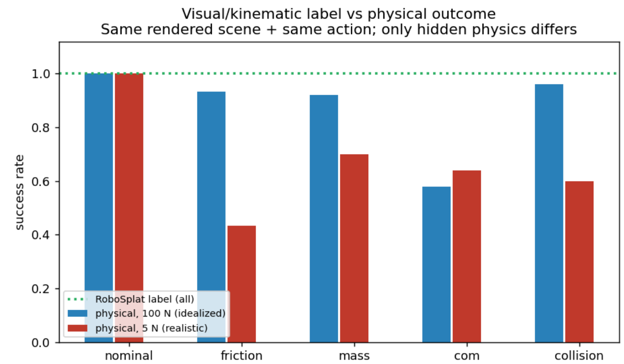

Figure 2. Same visual, different physical consequence. The RoboSplat-style visual/kinematic label remains 1.0 for all hidden-physics variants, while physical success diverges as hidden physical properties change.

Hidden property	Label	Physical success	Main failure
Nominal	1.0	1.00	–
Friction	1.0	0.43	slip / unstable lift
Mass	1.0	0.70	weak grasp
Center of mass	1.0	0.64	unstable lift
Collision width	1.0	0.60	unstable lift

Table 2. Same visual observation, different physical consequence. All groups share the same rendered observation, object pose, robot trajectory, and kinematic label; only hidden physical properties change during physical replay.

lift, each corresponding to a clear contact-dynamics mechanism. An idealized 100 N clamp can mask some friction, mass, and collision-width differences, while center-of-mass offset still causes substantial failure under strong grasping, showing that physics idealization itself affects verification conclusions.

Together, the pose-uncertainty and hidden-physics diagnostics expose a direct mismatch between visual/kinematic labels and physical task consequences. The observed failures follow structured physical mechanisms with stable nominal replay. This motivates a consequence-aware verifier that reasons over visible scene evidence, candidate action parameters, task predicates, and uncertainty in hidden physical properties.

5. Method

This section proposes the Physics-Aware Action-Effect Consistency Verifier (PAEC) for candidate action consequence verification in 3DGS robot digital twins. PAEC receives a scene representation S, task goal g, and candidate action a_i, and outputs a physical consequence consistency score, failure-type estimate, and optional action correction under low confidence. The central design choice is to explicitly separate visible state from hidden physical properties: visible state comes from 3DGS/RGB observations, object pose, and robot state, while friction, mass, center of mass, and contact geometry are represented as physical beliefs rather than directly determined from vision.

5.1. Overview

PAEC sits after generation or planning and before real execution. Candidate actions may come from RoboSplat-style demonstration generation, Gaussian world models, motion planners, vision-language-model planners, or visuomotor policies. Unlike modules that evaluate only rendering quality or trajectory reachability, PAEC estimates whether the action can realize the task predicate ϕ_g under hidden physical uncertainty.

Given a candidate action a_i, PAEC first extracts a visible proxy x_v from S and establishes a belief distribution $b(\theta)$ over hidden physical variables. The action encoder, robot-state encoder, and physical-belief encoder then produce features that are fused into a joint representation z_i through physics-conditioned fusion. The verifier outputs

$$(r_i,\, p_i^{\text{fail}},\, m_i,\, \Delta a_i) = \text{PAEC}(S, g, a_i, b(\theta)),$$

where r_i is the estimated probability that the action satisfies the task predicate, p_i^{fail} is the failure-type distribution, m_i contains task-related physical metrics such as lift height, slip distance, and tilt angle, and Δa_i is an optional action-correction residual.

5.2. Visible Proxy Construction

3DGS provides photorealistic rendering and spatial consistency, but its Gaussian primitives are not naturally equivalent to contactable physical objects. PAEC therefore first constructs a verification-oriented visible proxy:

$$x_v = h_{\text{proxy}}(S, g).$$

This proxy contains object pose, scale, visible geometric extent, approximate contact regions, candidate grasp regions, camera visibility, and task-relevant spatial relations. For grasping, x_v may include the object bounding box, relative position of candidate contact surfaces, object dimensions, and the relation between target lift direction and gripper closing direction. For pushing or placing, x_v may further include support surfaces, target regions, and obstacle relations.

The visible proxy provides state information supported by visual observations. It does not recover true friction, mass, or center of mass. This design avoids encoding visually unidentifiable physical properties as deterministic observations. In implementation, x_v can be encoded from structured geometric descriptors, Gaussian object features, RGB/depth features, or their combination, producing a visual/geometric embedding:

$$e_{\text{vis}} = E_{\text{vis}}(x_v).$$

5.3. Physical Belief Initialization

Let the hidden physical properties be

$$\theta = \{\mu, m, c, \kappa, F\},$$

where μ is friction, m is mass, c is center-of-mass offset, κ is contact or collision geometry, and F denotes grasp or control force conditions. The Same-Visual-Different-Physics diagnostic shows that identical visual observations can correspond to different θ values and different task consequences. PAEC therefore represents these hidden attributes with a belief distribution:

$$b_0(\theta) = p(\theta \mid x_v, \pi_{\text{cat}}, \pi_{\text{mat}}, \pi_{\text{task}}),$$

where π_{cat}, π_{mat}, and π_{task} denote category, material, and task priors.

In a minimal implementation, $b_0(\theta)$ can be given by manually specified physical ranges, such as friction range, mass range, upper bound on center-of-mass offset, and a set of candidate collision geometries. A fuller implementation can initialize the belief from object category, material cues, or historical interaction statistics. The key point is that PAEC does not need to know the true θ before manipulation; it evaluates whether the action is robust under the possible physical conditions covered by $b(\theta)$.

To input the belief into a network, we sample K physical hypotheses:

$$\theta_k \sim b(\theta), \quad k = 1, \ldots, K.$$

Each sample is encoded by a physical encoder:

$$e_{\theta,k} = E_\theta(\theta_k),$$

and then aggregated by mean pooling, attention pooling, or Set Transformer into a belief embedding:

$$e_{\text{phys}} = \text{Pool}\big(\{e_{\theta,k}\}_{k=1}^{K}\big).$$

This embedding describes physical uncertainty faced by the current action rather than a single physical state.

5.4. Candidate Action Encoding

The candidate action a_i can be represented as a low-level control sequence or as high-level action parameters. For grasping, we construct action features from

PAEC Pipeline Placeholder

3DGS/RGB observation + candidate action → visible proxy construction → physical belief initialization → physics-conditioned fusion → success / failure / metric / correction heads

Figure 3. PAEC overview placeholder. Given a 3DGS/RGB observation and a candidate action, PAEC first constructs a visible object proxy and initializes a belief distribution over hidden physical properties. Candidate action, robot state, and physical belief are encoded and fused through a physics-conditioned fusion module to predict task success probability, failure type, physical metrics, and action correction. Optional safe physical probing can update the physical belief before execution.

the end-effector trajectory, gripper width, grasp pose, approach direction, lift direction, lift speed, and grasp force. The action sequence is encoded as

$$e_{\text{act}} = E_{\text{act}}(a_i),$$

where E_{act} can be a temporal convolution, GRU, or Transformer encoder. The robot initial state and execution context are encoded as

$$e_{\text{robot}} = E_{\text{robot}}(q_0, x_{\text{ee}}, w_{\text{gripper}})$$

where q_0 is the initial joint state, x_{ee} is the end-effector pose, and w_{gripper} is the gripper state.

This action representation preserves the parameter structure needed for later correction. Treating an action as a black-box trajectory can yield a failure probability but makes it difficult to explain how the action should be repaired. Decomposing actions into grasp pose, force, width, and lift speed naturally links failure types to action adjustments.

5.5. Physics-Aware Verification

The key PAEC module is physics-conditioned fusion. Because the same action can produce different consequences under different physical properties, the physical belief should directly modulate the action representation. We use FiLM-style modulation as one implementation:

$$\gamma, \beta = M_{\text{film}}(e_{\text{phys}}),$$

$$\tilde{e}_{\text{act}} = \gamma \odot e_{\text{act}} + \beta.$$

The visible proxy, physically modulated action features, robot state, and physical belief are then fused:

$$z_i = M_{\text{fuse}}\big([e_{\text{vis}}, \tilde{e}_{\text{act}}, e_{\text{robot}}, e_{\text{phys}}]\big).$$

This fusion can also be implemented by cross-attention, allowing action-phase tokens to interact with friction, mass, center-of-mass, and contact-geometry tokens. The FiLM version is a compact implementation because it directly expresses that physical conditions change the executability of the same action.

The verifier predicts action consequences from z_i:

$$r_i = \sigma(h_{\text{succ}}(z_i)),$$

$$p_i^{\text{fail}} = \text{softmax}(h_{\text{fail}}(z_i)),$$

$$m_i = h_{\text{metric}}(z_i).$$

Here, r_i is task success probability; p_i^{fail} includes failure types such as weak grasp, slip, unstable lift, tilt/drop, and collision mismatch; and m_i regresses physical metrics such as maximum lift, slip distance, tilt angle, and contact duration.

With explicit sampled evaluation, the verifier can also predict for each physical hypothesis:

$$r_i = \frac{1}{K}\sum_{k=1}^{K} p_\psi(y = 1 \mid x_v, g, a_i, \theta_k),$$

and record a worst-case score:

$$r_i^{\text{worst}} = \min_k p_\psi(y = 1 \mid x_v, g, a_i, \theta_k).$$

The former measures average robustness, while the latter is useful for conservative action filtering in high-risk tasks.

5.6. Verifier-Guided Filtering and Correction

PAEC outputs are used for candidate action filtering and correction. The filtering form matches the problem formulation:

$$\mathcal{A}_{\text{valid}} = \{a_i \in \mathcal{A} \mid r_i > \tau\}.$$

When r_i falls below the threshold, the action may still be visually and kinematically plausible, but physically insufficiently robust. The system can discard the candidate or modify its parameters according to the predicted failure type.

The correction head outputs an action residual:

$$\Delta a_i = h_{\text{corr}}(z_i), \quad a_i' = a_i + \Delta a_i.$$

For grasping, Δa_i may adjust grasp force, gripper width, grasp point, lift speed, or lift direction. Different failure types correspond to different correction directions: slip risk often calls for higher grasp force or lower lift speed; weak grasp calls for gripper width, contact point, or force-margin adjustment; unstable lift or tilt calls for grasp-point and lift-direction adjustment.

Without a trained correction head, verifier-guided search can also be used:

$$a_i' = \arg \max_{\hat{a} \in \mathcal{N}(a_i)} V(S, g, \hat{a}),$$

where $\mathcal{N}(a_i)$ is a local parameter neighborhood around the original action, such as combinations of grasp force, gripper width, lift speed, and grasp offset. This search-based correction shares the same verifier interface as learned correction.

5.7. Optional Safe Physical Probing

For unseen objects, visual evidence and priors may not sufficiently constrain $b(\theta)$. PAEC can introduce low-risk physical probes before full execution, such as a light squeeze, micro-lift, gentle push, or small perturbation. The probe produces observation z_{probe}, including relative slip, tilt angle, gripper closure, contact duration, and micro-lift height change. The belief is updated by

$$b_1(\theta) = p(\theta \mid x_v, z_{\text{probe}})$$

and then used for subsequent verification and correction.

This module turns unknown object physics into test-time uncertainty updating, rather than requiring the visual model to recover friction, mass, or center of mass precisely before interaction. If probing is not allowed, PAEC can still perform conservative verification using a prior-only belief. If task risk or belief uncertainty is high, a probe-then-verify strategy is more suitable for reducing full-execution failure.

5.8. Training Signals

PAEC supervision comes from physical rollout rather than generation-time visual or kinematic labels. Each training sample contains a visible proxy, candidate action, robot state, physical hypothesis θ, and rollout consequence label. Consequence labels include task success, failure type, and continuous physical metrics. The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{succ}} + \lambda_f \mathcal{L}_{\text{fail}} + \lambda_m \mathcal{L}_{\text{metric}} + \lambda_c \mathcal{L}_{\text{corr}},$$

where $\mathcal{L}_{\text{succ}}$ is a binary success loss, $\mathcal{L}_{\text{fail}}$ is failure-type cross entropy, $\mathcal{L}_{\text{metric}}$ regresses physical metrics such as lift, slip, and tilt, and $\mathcal{L}_{\text{corr}}$ supervises the correction head when a search-derived corrected action exists.

This training design makes the network learn physical action effects rather than reproduce visual labels. In Same-Visual-Different-Physics samples with identical visual observations, RGB-only or kinematic-only inputs cannot distinguish different physical outcomes; PAEC learns consequence differences for the same action under different physical conditions through explicit physical beliefs and rollout-outcome supervision.

6. Experiments

7. Results

8. Discussion

9. Conclusion

References

[1] J. Abou-Chakra, K. Rana, F. Dayoub, and N. Sünderhauf. Physically embodied gaussian splatting: A realtime correctable world model for robotics. arXiv preprint arXiv:2406.10788, 2024. 3

[2] L. Barcellona, A. Zadaianchuk, D. Allegro, S. Papa, S. Ghidoni, and E. Gavves. Dream to manipulate: Compositional world models empowering robot imitation learning with imagination. arXiv preprint arXiv:2412.14957, 2024. 1, 3

[3] F. Ebert, C. Finn, S. Dasari, A. Xie, A. Lee, and S. Levine. Visual foresight: Model-based deep reinforcement learning for vision-based robotic control. arXiv preprint arXiv:1812.00568, 2018. 3

[4] A. Guédon and V. Lepetit. SuGaR: Surface-aligned gaussian splatting for efficient 3D mesh reconstruction and high-quality mesh rendering. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. 2

[5] X. Han, M. Liu, Y. Chen, J. Yu, X. Lyu, Y. Tian, B. Wang, W. Zhang, and J. Pang. Re^{3}Sim: Generating high-fidelity simulation data via 3D-photorealistic real-to-sim for robotic manipulation. arXiv preprint arXiv:2502.08645, 2025. 1, 2

[6] Y. Hu, Y. Guo, P. Wang, X. Chen, Y.-J. Wang, J. Zhang, K. Sreenath, C. Lu, and J. Chen. Video prediction policy: A generalist robot policy with

predictive visual representations. arXiv preprint arXiv:2412.14803, 2024. 3

[7] B. Huang, Z. Yu, A. Chen, A. Geiger, and S. Gao. 2D gaussian splatting for geometrically accurate radiance fields. In ACM SIGGRAPH Conference Papers, 2024. 2

[8] G. Jiang, H. Chang, R.-Z. Qiu, Y. Liang, M. Ji, J. Zhu, Z. Dong, X. Zou, and X. Wang. GSWorld: Closed-loop photo-realistic simulation suite for robotic manipulation. arXiv preprint arXiv:2510.20813, 2025. 1, 2

[9] D. Kalashnikov, A. Irpan, P. Pastor, J. Ibarz, A. Herzog, E. Jang, D. Quillen, E. Holly, M. Kalakrishnan, V. Vanhoucke, and S. Levine. QT-Opt: Scalable deep reinforcement learning for vision-based robotic manipulation. arXiv preprint arXiv:1806.10293, 2018. 3

[10] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3D Gaussian splatting for real-time radiance field rendering. ACM Transactions on Graphics, 42(4), 2023. 1, 2

[11] X. Li, J. Li, Z. Zhang, R. Zhang, F. Jia, T. Wang, H. Fan, K.-K. Tseng, and R. Wang. RoboGSim: A real2sim2real robotic gaussian splatting simulator. arXiv preprint arXiv:2411.11839, 2024. 1, 2

[12] F. Liu, H. Wang, S. Yao, S. Zhang, J. Zhou, and Y. Duan. Physics3D: Learning physical properties of 3D gaussians via video diffusion. arXiv preprint arXiv:2406.04338, 2024. 3

[13] G. Lu, B. Jia, P. Li, Y. Chen, Z. Wang, Y. Tang, and S. Huang. GWM: Towards scalable gaussian world models for robotic manipulation. In IEEE/CVF International Conference on Computer Vision (ICCV), 2025. 1, 3

[14] G. Lu, S. Zhang, Z. Wang, C. Liu, J. Lu, and Y. Tang. ManiGaussian: Dynamic gaussian splatting for multi-task robotic manipulation. arXiv preprint arXiv:2403.08321, 2024. 1, 3

[15] J. Mahler, J. Liang, S. Niyaz, M. Laskey, R. Doan, X. Liu, J. A. Ojea, and K. Goldberg. Dex-Net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics. In Robotics: Science and Systems (RSS), 2017. 3

[16] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In European Conference on Computer Vision (ECCV), 2020. 2

[17] K. Mo, L. Guibas, M. Mukadam, A. Gupta, and S. Tulsiani. Where2Act: From pixels to actions for articulated 3D objects. In IEEE/CVF International Conference on Computer Vision (ICCV), 2021. 3

[18] A. Mohtasib, A. Ghalamzan E., N. Bellotto, and H. Cuayáhuitl. Neural task success classifiers for robotic manipulation from few real demonstrations. arXiv preprint arXiv:2107.00722, 2021. 3

[19] B. Moran, M. Comi, A. Byravan, S. Bohez, T. Erez, Z. Li, and L. Hasenclever. Splatting physical scenes: End-to-end real-to-sim from imperfect robot data. arXiv preprint arXiv:2506.04120, 2025. 3

[20] T. Mu, Z. Ling, F. Xiang, D. Yang, X. Li, S. Tao, Z. Huang, Z. Xu, and H. Su. ManiSkill: Generalizable manipulation skill benchmark with large-scale demonstrations. In Advances in Neural Information Processing Systems Datasets and Benchmarks Track, 2021. 1, 2

[21] NVIDIA. NVIDIA PhysX SDK. https://github.com/NVIDIA-Omniverse/PhysX, 2025. Accessed 2026-06-16. 1, 2

[22] M. N. Qureshi, S. Garg, F. Yandun, D. Held, G. Kantor, and A. Silwal. SplatSim: Zero-shot sim2real transfer of RGB manipulation policies using gaussian splatting. arXiv preprint arXiv:2409.10161, 2024. 1, 2

[23] F. Xiang, Y. Qin, K. Mo, Y. Xia, H. Zhu, F. Liu, M. Liu, H. Jiang, Y. Yuan, H. Wang, L. Yi, A. X. Chang, L. J. Guibas, and H. Su. SAPIEN: A simulated part-based interactive environment. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020. 1, 2

[24] T. Xie, Z. Zong, Y. Qiu, X. Li, Y. Feng, Y. Yang, and C. Jiang. PhysGaussian: Physics-integrated 3D gaussians for generative dynamics. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. 3

[25] D. Xu, A. Mandlekar, R. Martín-Martín, Y. Zhu, S. Savarese, and L. Fei-Fei. Deep affordance foresight: Planning through what can be done in the future. arXiv preprint arXiv:2011.08424, 2020. 3

[26] X. Xu, W. Ge, D. Qiu, Z. Chen, D. Yan, Z. Liu, H. Zhao, H. Zhao, S. Zhang, J. Liang, and Y.-C. Chen. GaussianProperty: Integrating physical properties to 3D gaussians with LMMs. arXiv preprint arXiv:2412.11258, 2024. 3

[27] S. Yang, W. Yu, J. Zeng, J. Lv, K. Ren, C. Lu, D. Lin, and J. Pang. Novel demonstration generation with gaussian splatting enables robust one-shot manipulation. arXiv preprint arXiv:2504.13175, 2025. 1, 2, 4

[28] S. Zhou, Y. Du, J. Chen, Y. Li, D.-Y. Yeung, and C. Gan. RoboDreamer: Learning compositional world models for robot imagination. arXiv preprint arXiv:2404.12377, 2024. 3

[29] Z. Zhou and D. Negrut. ChronoDreamer: Action-conditioned world model as an online simulator for robotic planning. arXiv preprint arXiv:2512.18619, 2025. 3

[30] S. Zhu, G. Wang, X. Kong, D. Kong, and H. Wang. 3D gaussian splatting in robotics: A survey. arXiv preprint arXiv:2410.12262, 2024. 2